🧑‍🏫 **Teacher Copy — includes the Answer Key & Teacher Guide. Do not distribute to students.**

UNIT 8 · STANDARD 6.AT.8

Write and Solve Equations Using Addition or Subtraction

Lesson 8-2 · Teacher Copy

Name: Type your name **Date:** Today's date **Class:** Class period

My Notes Level 2 Standard

🎯 **TODAY'S OBJECTIVES**

Content Objective: I can solve one-step addition and subtraction equations using inverse operations.
Language Objective: I can explain my steps using the words equation, inverse operation, isolate, and solution.

📖 See **Learn It** for the explanation and worked examples, and the **Vocab** tab for the words. These are your notes to fill in and keep.

✏️ **Fill in each blank as we go. Use the Word Bank to help you.**

📚 **WORD BANK — FILL EACH BLANK WITH THE BEST WORD**

Solve One-Step Addition and Subtraction Equations | Equation | Inverse operation | Isolate | Solution | One-step equation | Balance

👆 *Tap any word to see what it means and a picture.*

1. Undo an addition by ____________ the same amount from both sides.

2. A math sentence with an equal sign showing two equal amounts is an ____________.

3. An operation that undoes another, like subtraction undoing addition, is an ____________.

4. To get the variable alone on one side of the equation is to ____________ it.

5. The value of the variable that makes the equation true is the

______________________.

6. An equation solved in a single step is a ______________________.

7. Doing the same thing to both sides to keep them equal keeps the equation in

______________________.

Write About the Math

Guided ESOL writing

Use the support level you need. Every level answers the same math question.

1. Understand the Question

EXPLAIN

How far has the hiker walked? How did you use inverse operations?

Your job: Answer the question. Use math evidence. Explain how the evidence proves your answer.

Responde la pregunta. Usa evidencia matemática. Explica cómo la evidencia demuestra tu respuesta.

2. Plan Your Math Words

Check at least two words you will use.

- [ ] **Solve One-Step Addition and Subtraction Equations** — Use one inverse addition or subtraction operation to isolate the variable.
  Resolver ecuaciones de suma y resta de un paso — Usar una operación inversa de suma o resta para aislar la variable.
- [ ] **Equation** — A math sentence with an equal sign showing both sides are the same.
  Ecuación — Una oración matemática con un signo igual que muestra que ambos lados son iguales.
- [ ] **Inverse operation** — Two math actions that undo each other, like × and ÷.
  Operación inversa — Dos operaciones que se deshacen entre sí, como × y ÷.
- [ ] **Isolate** — To get the letter by itself on one side.
  Despejar — Dejar la letra sola en un lado.
- [ ] **Solution** — A number that makes the equation or inequality true.
  Solución — Un número que hace verdadera la ecuación o desigualdad.

Say it first: Say your idea to a partner or quietly to yourself before you write.
Di tu idea a un compañero o en voz baja antes de escribir.

The hiker walked ____ miles because d + 4.5 = 12 means d = 12 − ____ = ____.

Di tu respuesta primero: Mi respuesta es ____ porque ____.

3. Build Your Explanation

Model the parts: This model shows the parts of an explanation. It does not answer your writing question.

Claim: I can undo the addition to get the variable by itself.

Evidence: Students recognize 23 is added to n and that subtracting 23 (the inverse operation) from both sides will isolate n.

Reasoning: This evidence connects the definition of equation to the mathematical claim.

Start

Most language support

Complete the frames. Use the word bank.

Completa las oraciones. Usa el banco de palabras.

- **The hiker walked ____ miles because d + 4.5 = 12 means d = 12 − ____ = ____.**
- **My answer is ____.**
  Mi respuesta es ____.

Build

Some language support

Write two connected sentences. Use because, so, or but.

Escribe dos oraciones conectadas. Usa porque, entonces o pero.

- **My claim is ____.**
  Mi afirmación es ____.
- **I know because ____.**
  Lo sé porque ____.
- **So ____.**
  Entonces ____.

Explain

Light language support

Write a claim, give math evidence, and explain your reasoning.
Escribe una afirmación, da evidencia matemática y explica tu razonamiento.

- **My claim is ____.**
  Mi afirmación es ____.
- **My math evidence is ____.**
  Mi evidencia matemática es ____.
- **This proves ____ because ____.**
  Esto demuestra ____ porque ____.

4. Check Your Explanation

- [ ] I answered the question.
  Respondí la pregunta.
- [ ] I used math evidence: numbers, an equation, a model, or a comparison.
  Usé evidencia matemática: números, una ecuación, un modelo o una comparación.
- [ ] I explained how the evidence proves my answer.
  Expliqué cómo la evidencia demuestra mi respuesta.
- [ ] I used at least two math words.
  Usé por lo menos dos palabras matemáticas.
- [ ] I reread complete sentences and punctuation.
  Volví a leer mis oraciones completas y la puntuación.

Writing Guide — Teacher Copy

The support level changes the amount of language scaffolding, not the mathematical expectation. Look for the same five features in every response.

- I answered the question.
- I used math evidence: numbers, an equation, a model, or a comparison.
- I explained how the evidence proves my answer.
- I used at least two math words.
- I reread complete sentences and punctuation.

Answer Key & Teacher Guide

1. **Notes 1:** Solve One-Step Addition and Subtraction Equations
2. **Notes 2:** Equation
3. **Notes 3:** Inverse operation
4. **Notes 4:** Isolate
5. **Notes 5:** Solution
6. **Notes 6:** One-step equation
7. **Notes 7:** Balance
8. **Watch for this mistake:** *A common mistake is applying the same operation instead of the inverse — for example, solving $x - 7 = 15$ by subtracting 7 from both sides (getting the wrong $x = 8$) instead of adding 7 to both sides (the correct $x = 22$). Another common mistake is doing the inverse operation to only one side, like solving $x + 6 = 14$ by subtracting 6 from the left but forgetting the right side. Before you submit, name the operation shown in the equation, choose its inverse, apply it to BOTH sides, and check by substituting your answer back into the original equation.*
9. **Try It 1:** A. $n = 13$ — *Add 4 to both sides: $n = 9 + 4 = 13$. Check: $13 - 4 = 9$ ✓*
10. **Try It 2:** A. 34 — *Subtract 16 from both sides: $t = 50 - 16 = 34$. Check: $34 + 16 = 50$ ✓*